

Utkarsh Singh

Software engineer

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Professional experience

Jaguar Land Rover Software Engineer

Jul 2022 – Oct 2025, Bengaluru

Vehicle network simulator

- Developed the in-house simulator in **Python** with a 3-engineer team, replacing Intrepid, Vector & dSPACE tool licences worth a projected **£500k**.
- Developed desktop, headless & touchscreen versions using **PySide6, Flask & reTerminal** (Raspberry Pi), used on **500+ machines** per telemetry.
- Built XML parser using **ElementTree** by reverse engineering the undocumented **100k-line spec**, rarely needing changes for quarterly releases.
- Implemented a **C++** port of the parser using pugixml & std::map of typed structs for real-time margins, cutting the parse time to **0.1s from 2-3s**.
- Solely owned the signal encoding, decoding & transmission end-to-end using **scapy & threading**, shipping **20+ features** such as fault injection.
- Built dashboard builder in **PySide6** with custom **drag-and-drop** widgets & per-widget code editor, exporting dashboards as a standalone .py file.
- Developed the touchscreen app for **100+ Apple CarPlay lab rigs** using **reTerminal**, with 10 signals such as vehicle speed, gear, wakeup & sleep.

Data engineering

- Built **Jira-to-BigQuery** data pipelines for **25+ projects** using Python's jira library, Cloud Functions & Cloud Scheduler, running 2-hourly to weekly.
- Designed **40+ Tableau dashboards** reading from BigQuery for management to self-serve, ranked **#1** among all creators org-wide based on usage.
- Developed **Selenium** scraper to extract XML data from 20+ internal REST endpoints into BigQuery tables covering all **40k** JLR employee records.

Workflow automation

- Built **Excel-to-Jira** automation in PySide6 for weekly supplier sheets, creating **1000s of tickets** & colour-coding mismatches in the sheet itself.

Independent projects

gitsearch [demo](#), [code](#)

Apr – Aug 2026

- Developed semantic search over **244k GitHub repos** with 200+ stars, using Python, FastAPI & **Postgres + pgvector**, finding them by what they do.
- Utilised GitHub GraphQL & REST APIs to crawl **267k repos** & index 244k with **bge-small** embeddings, with weekly refresh jobs on GitHub Actions.
- Implemented **hybrid retrieval** using Postgres full text, pgvector & pg_trgm in a **single SQL query**, with a breakdown of why each result ranked.
- Built an AI-powered guide agent using **Claude tool calls** over the repo files, allowing users to get a step-by-step usage guide for any search result.
- Utilised Supabase, GCP Cloud Run & GitHub Pages to host the full stack under **\$30/month** with scale-to-zero, serving warm searches in **120ms**.

The Good Guest [site](#), [code](#)

Jan – Aug 2026

- Designed & developed a product spanning an **iOS capture app, GCP backend & 3D web viewer**, letting users scan a room & redesign it over chat.
- Built capture flow in **SwiftUI & ARKit** with **RoomPlan** for room geometry, uploading the LiDAR scans to Google Cloud Storage in the background.
- Utilised Meta's **SAM 3 & SAM 3D Objects** for detecting & rebuilding furniture in 3D such as sofas, beds & tables in FastAPI on Cloud Run L4 GPUs.
- Developed a 3D room viewer in **Next.js, three.js & WebGL2** for exploring the scanned rooms in any browser, with **5.8x smaller** splat files via SPZ.
- Built an AI redesign agent using **Claude tool calls** with a backend solver checking each move, allowing users to rearrange furniture by chatting.
- Utilised **Firebase Auth SDK** for Apple & Google sign-in, creating iOS flows such as Live Activity progress & account deletion, as per App Store rules.

Smaller builds

Jul – Aug 2026

- Built **Shrink to Fit** for compressing images & PDFs to exact sizes like 500KB or 2MB in the browser using canvas, pdf.js & **Ghostscript WASM**. [code](#)
- Built read-only chat viewers for **WhatsApp & Instagram** exports in **vanilla JS** with voice note waveforms, search, zip repair & **zero uploads**. [code](#)
- Built 20 generative art pieces such as harmonographs, Lissajous tables & Truchet tiles in **HTML Canvas & GLSL** for linework, exporting plotter SVG.

Education

Indian Institute of Technology Delhi B.Tech + M.Tech Dual Degree, Chemical Engineering, CGPA 8.6/10

Jul 2017 – May 2022

- Built **packed-bed reactor geometries** in **Blender's** rigid body physics for the thesis, with porosity analysis in **MATLAB** & CFD runs in **Ansys Fluent**.

Skills

AI engineering

AI Agents | Tool Use | Anthropic API | Gemini API | Embeddings | pgvector | Hybrid Search | LLM Evals | Gaussian Splatting

Backend

Python | C++ | FastAPI | Flask | PostgreSQL | SQL | Protobuf | scapy | Multithreading | Docker | GitHub Actions

Frontend & mobile

Swift | SwiftUI | ARKit | RoomPlan | Next.js | React | TypeScript | JavaScript | three.js | WebGL | GLSL | PySide6 | Qt

Cloud

GCP | Cloud Run | Cloud Functions | Cloud Scheduler | BigQuery | Firestore | Firebase | Supabase | Embedded Linux

Data

BigQuery | Tableau | pandas | Selenium | Jira

Honors & awards

- **Exceptional Creator Award**, Jaguar Land Rover, 2025
- **Department Rank 1**, Chemical Engineering Dual Degree batch, IIT Delhi, 2022
- Semester Merit Award ×2, IIT Delhi
- JEE Advanced AIR 2525 (**top 1.14%**), JEE Main AIR 742 (**top 0.06%**), 2017
- SOF Mathematics Olympiad, Zonal Rank 3 (finals); Rank 3, inter-school mathematics exam, awarded by the **Governor of Uttar Pradesh**